

Span

Kuch vectors se kitni jagah 'reach' ki ja sakti hai

01 Opening Hook

STORY TIME

Socho tumhare paas sirf 2 tarah ke Lego blocks hain – ek 1-unit lamba red block, ek 1-unit lamba blue block, dono perpendicular directions mein. In dono blocks ko combine karke (kitne bhi copies use karke) tum poore 2D grid ke kisi bhi point tak pahunch sakte ho. Yeh 'saari possible reachable jagah' ka naam hai ****Span****.

02 Why Does This Concept Exist?

THE PROBLEM

Agar humein jaanna hai ki kuch vectors se hum kya-kya bana (represent kar) sakte hain, humein span ka concept chahiye. Yeh batata hai ki given vectors se kitni 'coverage' mil sakti hai – kya woh poore space ko cover karte hain, ya sirf ek line/plane ko?

03 Explain Like I'm 10

SIMPLE WORDS

Socho tumhare paas do paint colors hain – red aur blue. Unhe alag-alag amounts mein mix karke tum bahut saare shades bana sakte ho (purple, light purple, dark red-purple, etc.) – yeh saare possible mixtures 'span' hain in do colors ka.

Agar tumhare paas sirf ek color hai (sirf red), tum sirf uske shades bana sakte ho (light red se dark red tak) – yeh ek chhota span hai, poora color-space cover nahi karta.

04 Build Intuition First (No Maths Yet)

FEEL IT FIRST

Span ek ya zyada vectors ke saare 'linear combinations' (scaling + adding) se banne wali saari jagah hai. Agar 2 independent vectors ka span poora 2D plane cover karta hai, matlab woh vectors 'kaafi' hain us space ko describe karne ke liye.

05 Ab Thoda Maths

FORMULA TIME

Definition:

$\text{Span}\{v_1, v_2, \dots, v_n\}$ = saare possible $c_1*v_1 + c_2*v_2 + \dots + c_n*v_n$, jahan $c_1, \dots, c_n$ koi bhi real numbers ho sakte hain.

Example:

Agar $v_1=[1,0]$ aur $v_2=[0,1]$, toh $\text{span}\{v_1, v_2\}$ = poora $\mathbb{R}^2$ (kyunki koi bhi (x,y) point $x*v_1 + y*v_2$ se ban sakta hai).

06 Visualize Everything

PICTURE IT

Span of ONE vector $v=[1,1]$:
(sirf ek line)

/

Span of TWO independent vectors:
(poora 2D plane)

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07 AI Connection

WHERE AI USES THIS

Feature Representation: Neural network ke ek layer ke neurons ka span batata hai ki woh layer kya-kya representations bana sakti hai – agar span 'chhota' hai, layer limited patterns hi seekh sakti hai.

Basis of Embedding Space: Embedding dimensions ka span poori 'meaning space' define karta hai jisme concepts represent ho sakte hain.

08 Real Industry Example

REAL COMPANIES

Research (Neural Network Capacity): Researchers layer ke weight matrix ke span/rank ko analyze karte hain yeh samajhne ke liye ki network kitna 'expressive' hai.

Signal Processing companies: Audio/video compression mein span ka concept use hota hai yeh decide karne ke liye ki kitne 'basis signals' se original signal accurately represent ho sakta hai.

09 Explain Like I'm Going To Build It

HOW YOU'D CODE IT

Practically, span ko directly code nahi karte, lekin iska concept `torch.linalg.matrix_rank(A)` se related hai – yeh batata hai matrix ke columns kitni 'independent directions' span karte hain, jo neural network capacity analysis mein useful hota hai.

10 Common Misconceptions

BEGINNER MISTAKES

Galat soch: Zyada vectors ka span hamesha bada hota hai.

Sahi baat: Agar naye vectors purane vectors ke span mein already hain (redundant hain), span badhta nahi – span sirf 'naye independent directions' add karne se badhta hai.

Galat soch: Span aur set of vectors ek jaisa hota hai.

Sahi baat: Span un vectors se ban sakne wali saari possible combinations hai, jo original set se bahut bada (infinite) ho sakta hai.

11 Interview Questions

INTERVIEW CORNER

[Beginner] Q: Span kya hai?

A: Kuch diye gaye vectors ke saare possible linear combinations (scaling + adding) se banne wali jagah.

[Intermediate] Q: Do vectors poore 2D plane ko span kar sakte hain kya condition mein?

A: Agar dono vectors linearly independent hain (ek doosre ka scalar multiple nahi hain), tabhi woh poore 2D plane ko span kar sakte hain.

[Advanced] Q: Neural network layer ka span uski 'expressiveness' se kaise related hai?

A: Agar layer ke weight vectors ka span chhota hai (low rank), layer sirf limited patterns represent kar sakti hai – iska matlab hai model underfitting ka risk hai us layer ki wajah se.

12 Exercises

PRACTICE TIME

[Easy] Ek single vector $[2,0]$ ka span kya hoga?

Solution: X-axis ki poori line (saare points jo $[2,0]$ ke scalar multiples hain).

[Medium] Kya $v_1=[1,2]$ aur $v_2=[2,4]$ poore R^2 ko span karte hain?

Solution: Nahi, kyunki $v_2 = 2 \cdot v_1$ (dependent vectors hain), unka span sirf ek line hai, poora plane nahi.

[Hard] 3 vectors R^3 mein hain lekin unka span sirf ek plane (2D) hai. Iska kya matlab hai?

Solution: Matlab teenon vectors linearly dependent hain - ek vector doosre don ka combination hai, isliye woh poore 3D space ko cover nahi kar paate, sirf ek 2D plane cover karte hain.

13 Golden Summary

Concept	Easy Meaning
Span	Vectors ke saare possible linear combinations se bani jagah
Span of 1 vector	Ek line
Span of 2 independent vectors (2D)	Poora plane
Redundant Vector	Jo span ko nahi badhata, kyunki already covered hai
Neural Network relevance	Layer capacity/expressiveness se related

14 One-Line Memory Trick

“Span batata hai diye gaye vectors se tum kahan-kahan 'pahunch' sakte ho.”